Shape of data

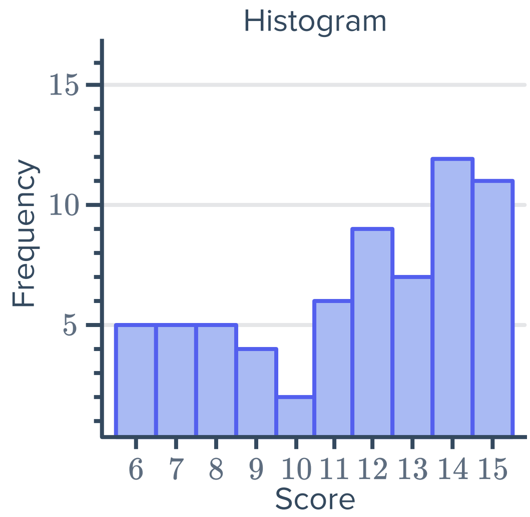
1 Describe the shape of the data in the following graphs:

a

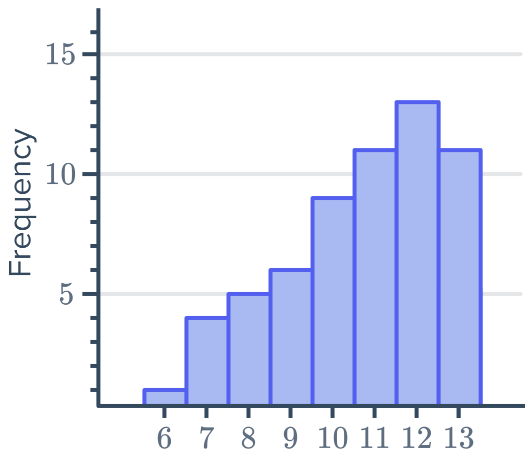
	Leaf
1	6 7 7
2	2 2 2 2 3 3 3
3	3 3 3 6 6 6 7 7 7 7 7
4	4 4 4 4 4 4
5	7 7

Key: $2|3 = 23$

b



c



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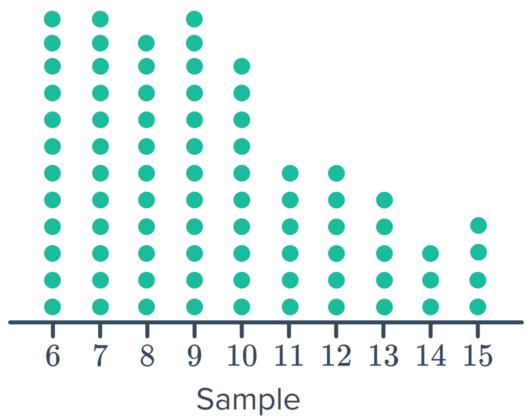
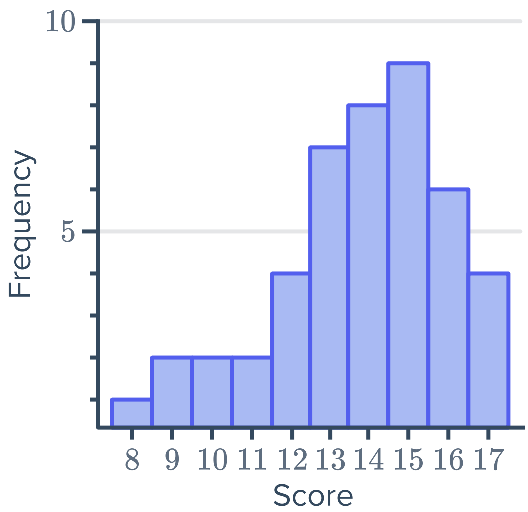


e



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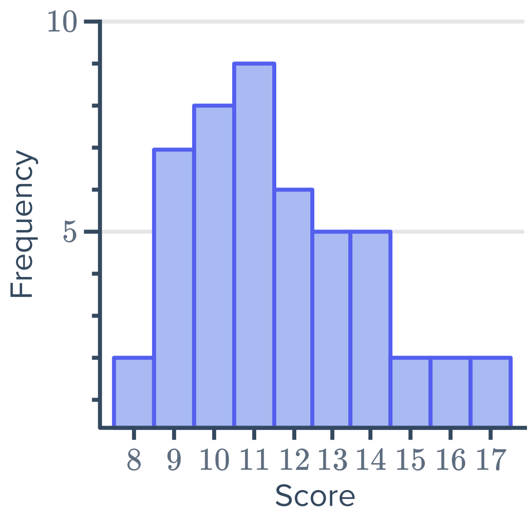

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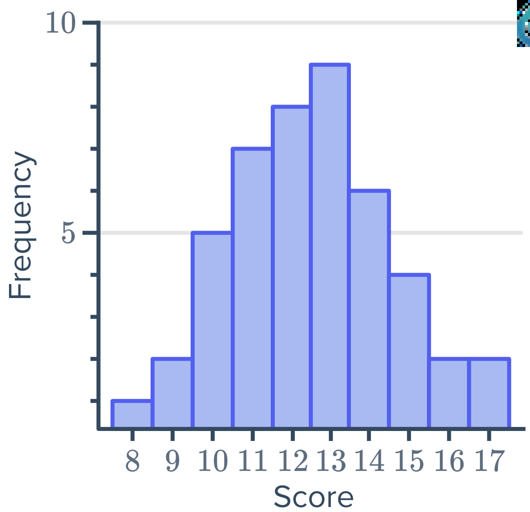
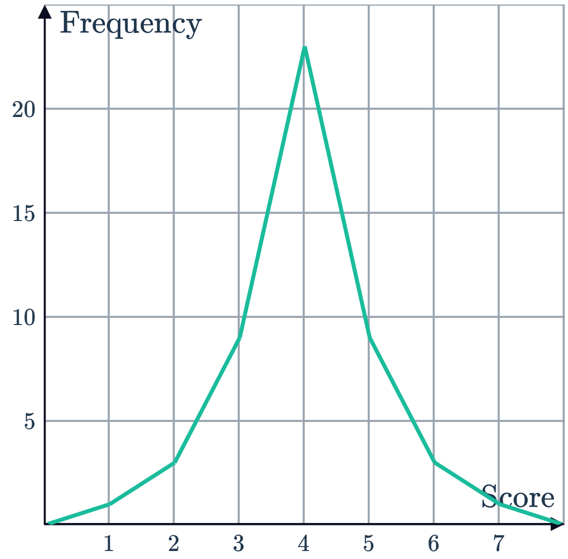
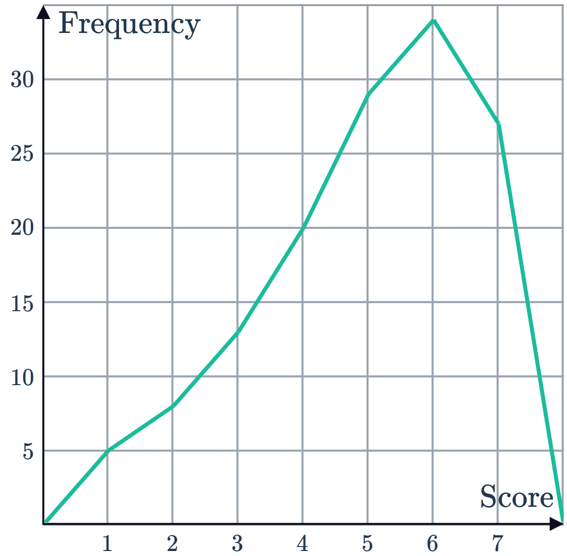


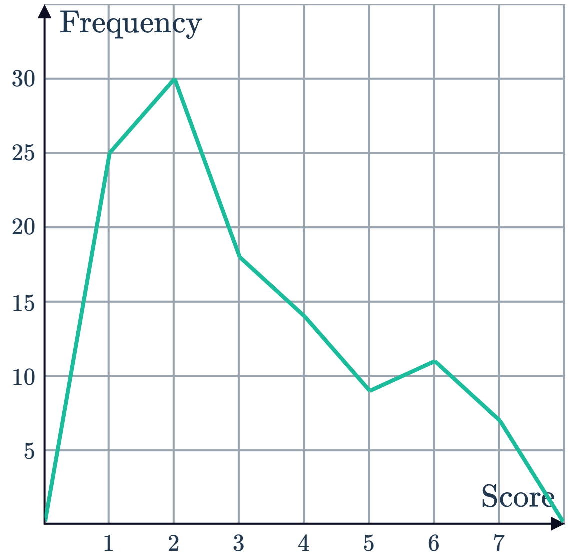
h



i



j



2 Consider the given column graph:

a Describe the shape of the distribution.

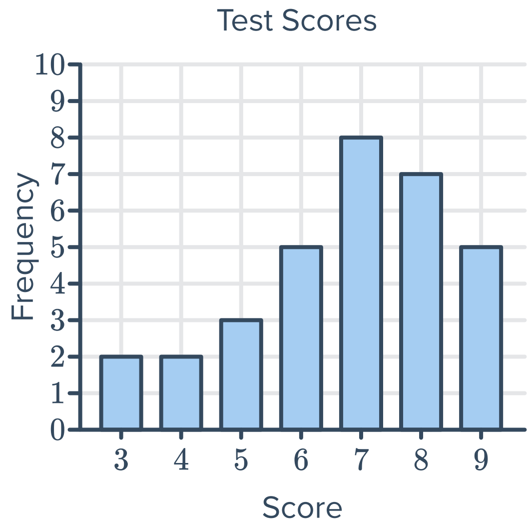
b Find the following:

i Lower quartile

ii Upper quartile

c Calculate the interquartile range.

d Are there any outliers? If so, state the value.



3 Consider the dot plot given:

a Describe the shape of the distribution.

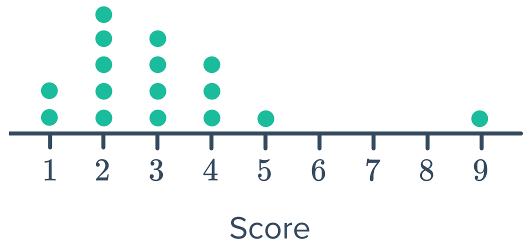
b Find the following:

i Lower quartile

ii Upper quartile

c Calculate the interquartile range.

d Are there any outliers? If so, state the value.



4 The stem-and-leaf plot below shows the age of people to enter through the gates of a concert in the first 5 seconds:

a Find the median age.

b Find the difference between the least age and the median.

c Find the difference between the greatest age and the median.

d Calculate the mean age, correct to two decimal places.

e Is the data positively or negatively skewed?

	Leaf
1	0 1 2 2 3 3 4 4 4 8 8 8
2	1 7
3	4 5 5
4	0
5	4

Key: $1|2 = 12$ years old

5 If a set of data is strongly positively skewed and the median is 70, what can we conclude about the mean?

6 A pair of dice are rolled and the numbers appearing on the uppermost face are added to create a score.

a How many combinations would result in a score of:

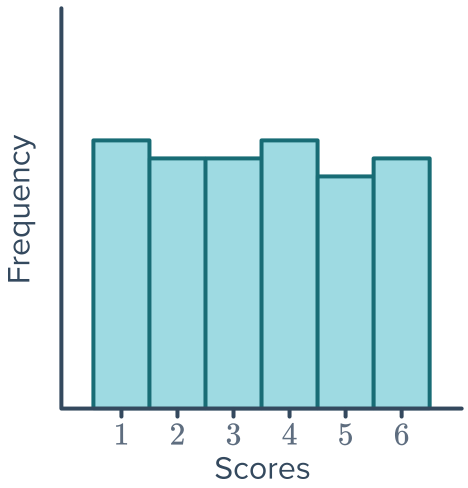
i 4 **ii** 7 **iii** 10

b A pair of dice are rolled for a large number of trials and the numbers appearing are added to create a score. Draw a sketch of a histogram you would expect to match the data.

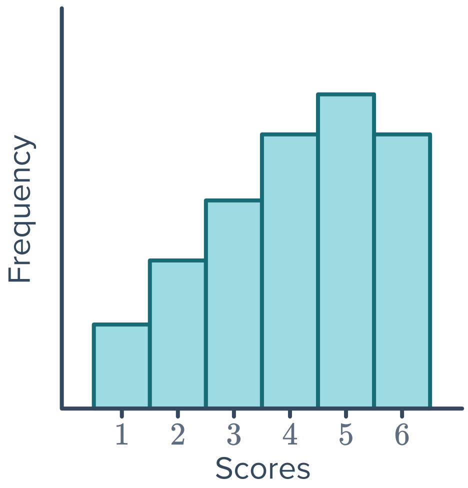
7 A die is rolled for a large number of trials and the number appearing is noted.

a Which histogram would you expect to match the data? Explain your answer.

A



B

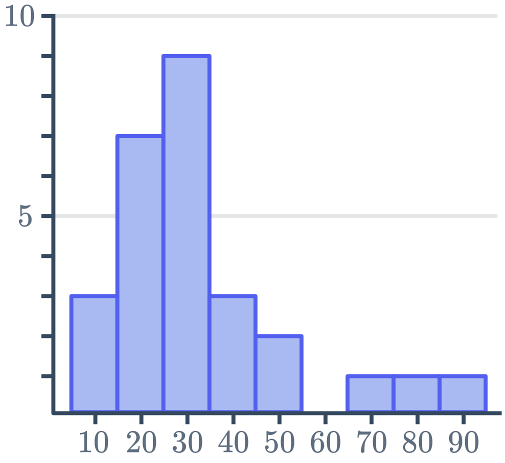


b Describe the shape of the distribution of the data chosen above.

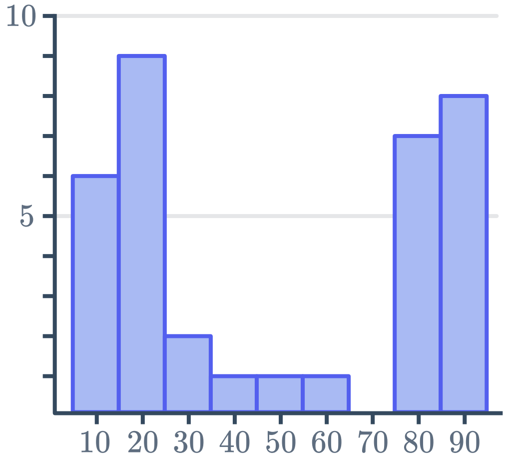
Box plots and histograms

8 Construct a box plot for the following histograms:

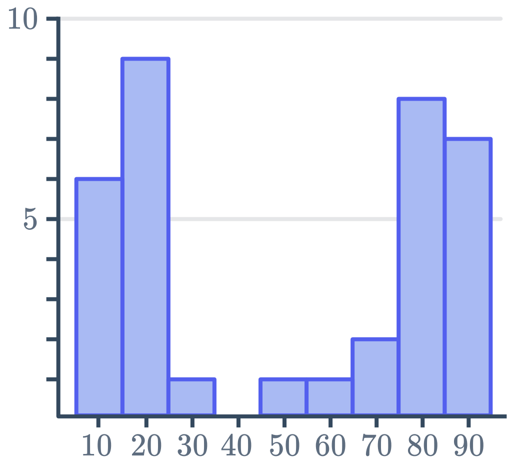
a



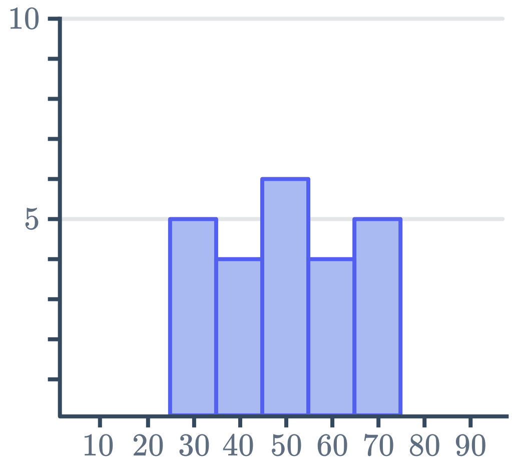
b



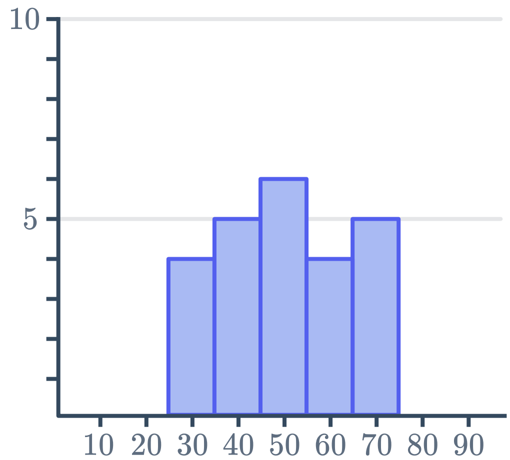
c


10
5
10 20 30 40 50 60 70 80 90

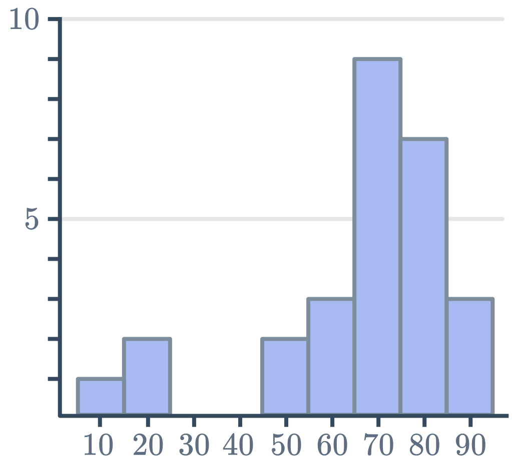
d


10
5
10 20 30 40 50 60 70 80 90

e

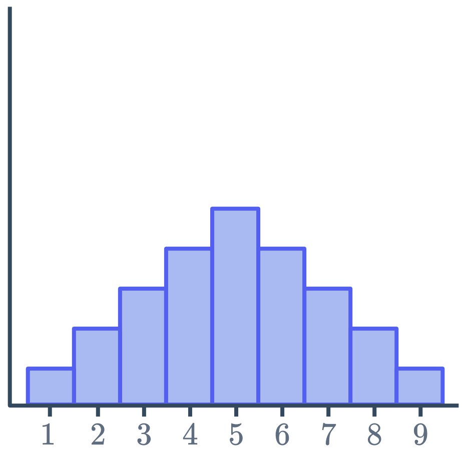

10
5
10 20 30 40 50 60 70 80 90

f

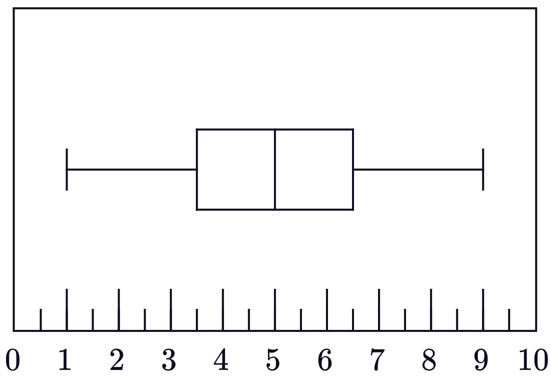

10
5
10 20 30 40 50 60 70 80 90

9 Match the histograms on the left to the corresponding box plots on the right:

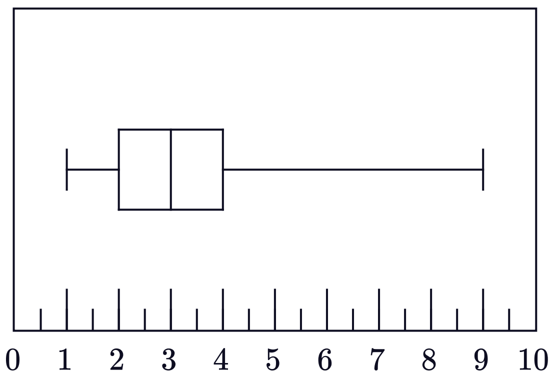
Histogram A



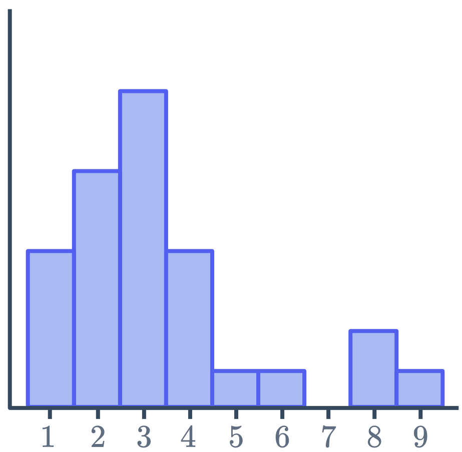
Box Plot 1



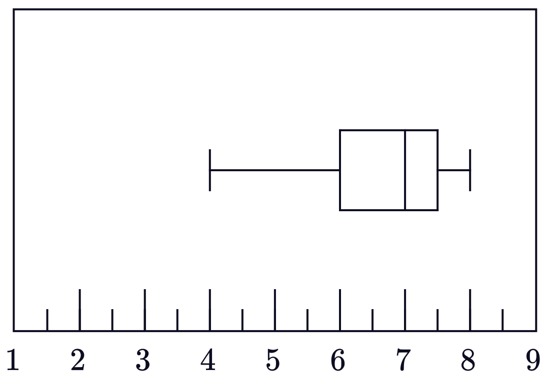
Box Plot 2



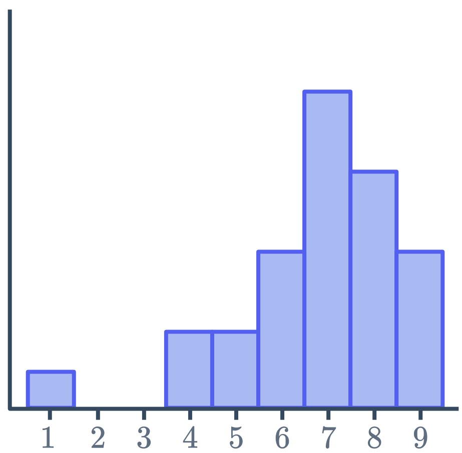
Histogram B



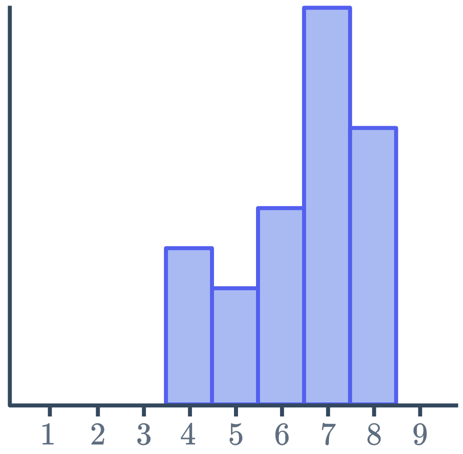
Box Plot 3



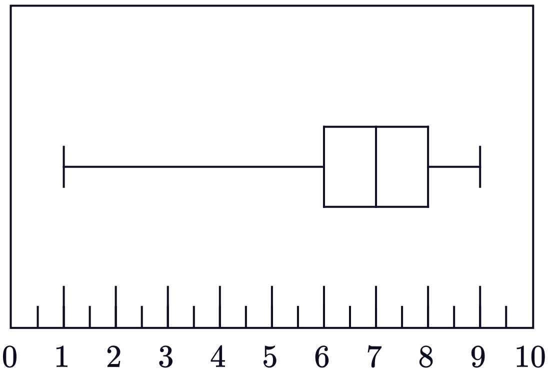
Histogram C



Histogram D

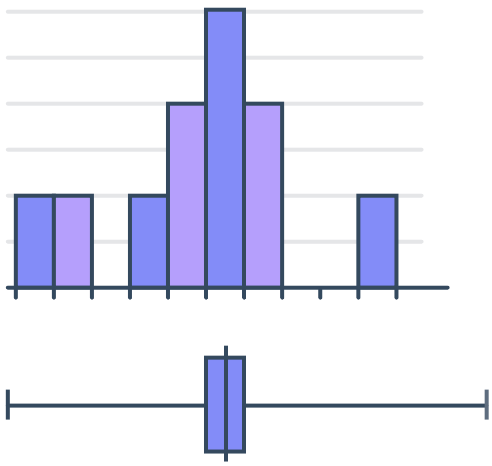


Box Plot 4

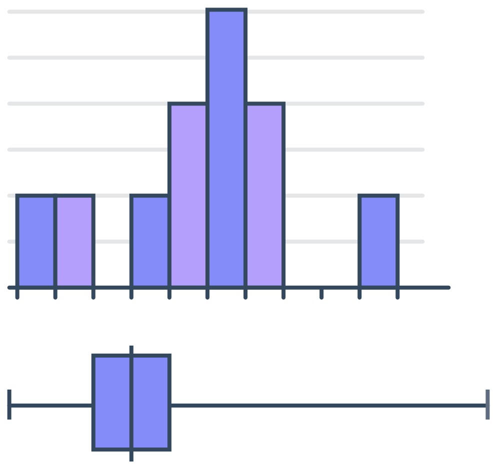


10 State whether the following pairs of histograms and box plots match with respect to their shape:

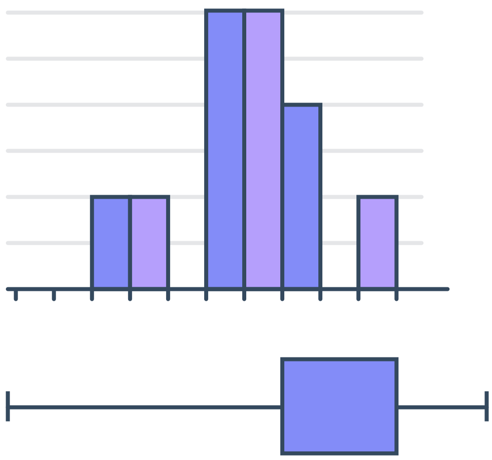
a



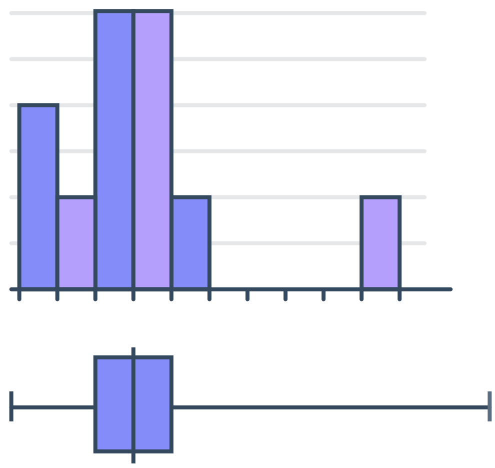
b



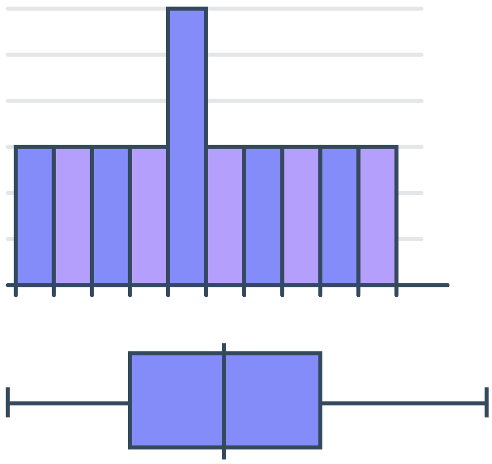
c



d



e



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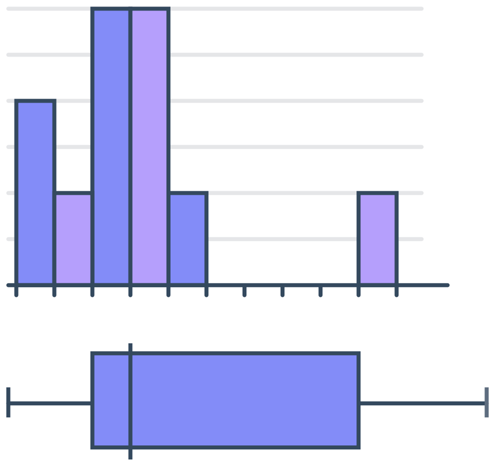
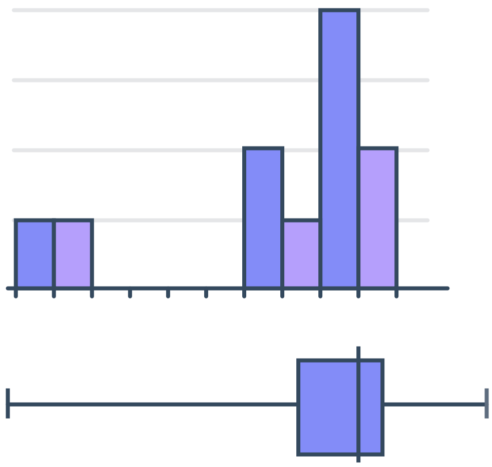
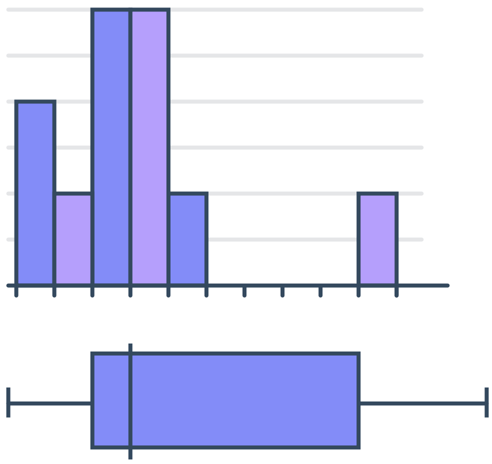


11 Explain why the following pairs of histograms and box plots do not match:

a



b



Measure of center

12 Which measure of center would be best for the following data set?

a $12, 15, 16, 21, 22, 25$

b $15, 13, 16, 17, 15, 15, 15$

c $8, 10, 14, 18, 19, 91$

13 State the most appropriate measure of cer[redacted]r the following scenarios:

a For a particular street, house prices for houses sold in the last 36 months (in 1000's) were given:

$$920,\ 920,\ 709,\ 694,\ 667,\ 657,\ 630,\ 610,\ 584,\ 567,\ 555$$

b The number of animal races that were won by a trainer over the years shown are listed in the table:

Year	2003	2004	2005	2006	2007
Races won	116	105	102	108	113

c The number of minutes spent studying by each member of a group of students is given:

$$85,\ 85,\ 95,\ 70,\ 60,\ 70,\ 105$$

d A netball team has so far played 8 games in the season. Their final score in each game is given:

$$36,\ 46,\ 49,\ 40,\ 52,\ 44,\ 39,\ 37$$

14 The selling price of recently sold houses is given below:

$$\$760\,000,\ \$650\,000,\ \$810\,000,\ \$780\,000,\ \$760\,000,\ \$590\,000,\ \$1\,360\,000$$

a Find the mean selling price. Round your answer to the nearest thousand dollars.

b Find the median selling price.

c Recalculate the mean selling price excluding the outlier.

d Recalculate the median selling price excluding the outlier.

e Which measure of center best identifies the typical selling price of recently sold houses? Explain your answer.

15 The weight of fish caught in a "weigh and release" fishing competition, in kilograms are given below:

$$12.5,\ 15.1,\ 13,\ 14.2,\ 14.5,\ 14.9,\ 12.5,\ 14.3,\ 1.5$$

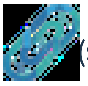
a Find the mean weight.

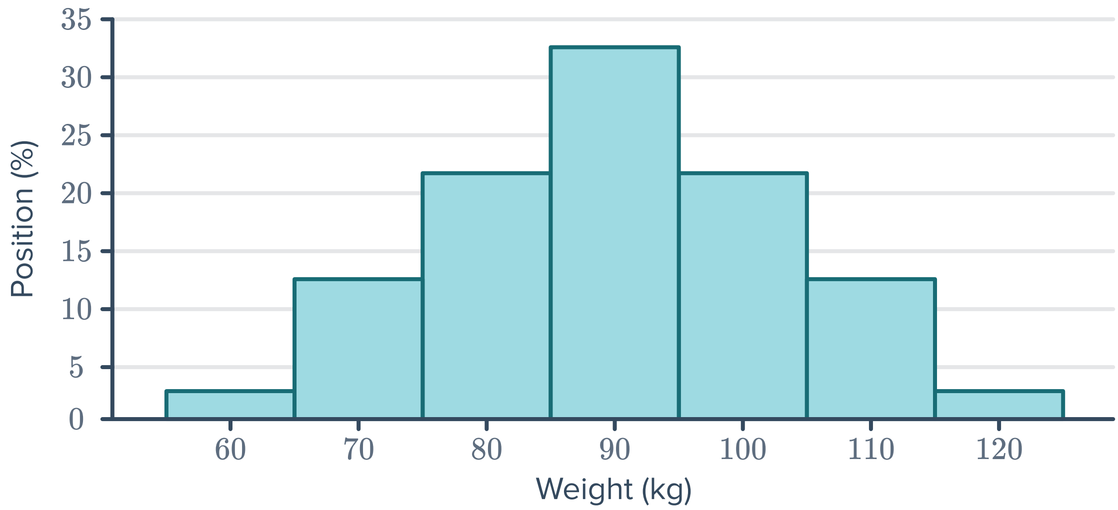
b Find the median weight.

c Recalculate the mean weight excluding the outlier.

d Recalculate the median weight excluding the outlier.

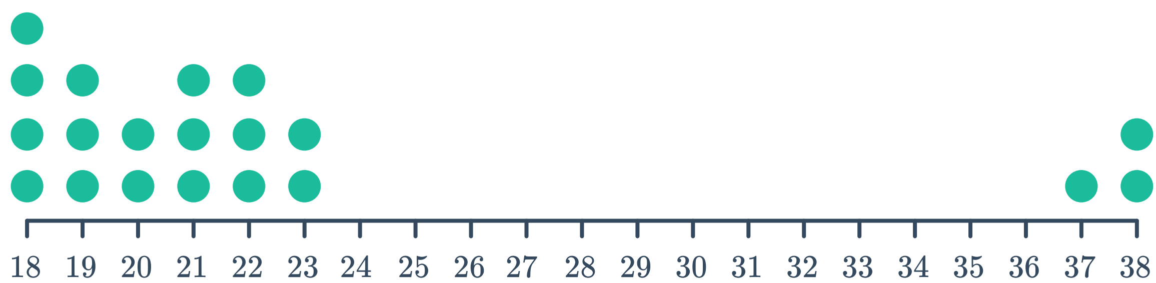
e Which measure of center best identifies the typical fish weight? Explain your answer.

16 Consider the histogram shown. Which me(s) of center best represents the center of the data set? Explain your answer.



17 The salaries of part-time employees at a company are given in the dot plot below. Which measure of center best reflects the typical wage of a part-time employee? Explain your answer.

Salaries (in $1000's)



18 A journalist wanted to report on road speed cameras being used as revenue raisers. She obtained data that showed the number of times 20 speed cameras issued a fine to motorists in one month. The results were:

$$101, 102, 115, 115, 121, 124, 127, 128, 130, 130,$$
$$143, 143, 146, 162, 162, 163, 178, 183, 194, 977$$

The journalist wants to give the impression that speed cameras are just being used to raise revenue. Which measure of center should she use in her article? Explain your answer.

Boxplots and outliers

19 For each of the following data sets, calculate:

i The interquartile range

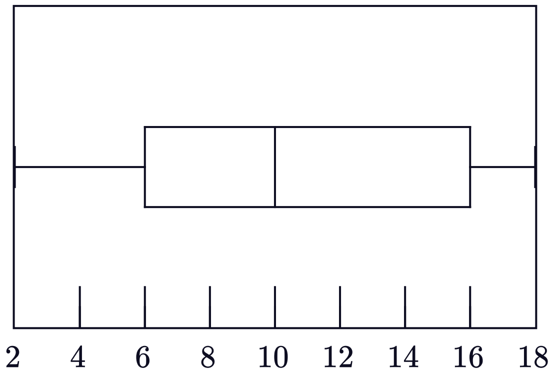
ii The value of the lower fence

iii The value of the upper fence

a

Minimum	5
Q1	6
Median	12
Q3	17
Maximum	28

b



20 For each of the following sets of data:

i Construct the five-number summary.

ii Calculate the interquartile range.

iii Calculate the value of the lower fence.

iv Calculate the value of the upper fence.

v Would the value -5 be considered an outlier?

vi Would the value 16 be considered an outlier?

a $9, 5, 3, 2, 6, 1$

b $3, 10, 9, 2, 7, 5, 6$

c $12, 5, 11, 1, 9, 8, 5, 6$

21 For each of the following sets of data:

i Construct the five-number summary.

ii Would the value -3 be considered an outlier?

iii Would the value 15 be considered an outlier?

a $1, 4, 8, 10, 6, 2, 5$

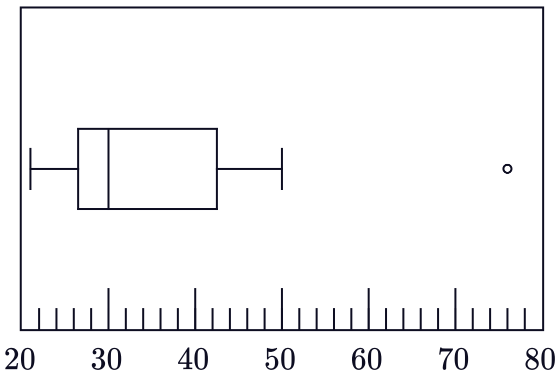
b $9, 4, 6, 11, 10, 8, 10$

22 The data point 5 is below the lower fence and is considered an outlier. The interquartile range is 12.
Find n, the smallest integer value the lower quartile can be.

23 The data point 37 is above the upper fenc[illegible] is considered an outlier. The interquartile range is 10.
Find n, the largest integer value the upper quartile can be.

24 $\mathrm{VO_2Max}$ is a measure of how efficiently your body uses oxygen during exercise. The more physically fit you are, the greater your $\mathrm{VO_2Max}$. Here are some people's results, listed in ascending order, when their $\mathrm{VO_2Max}$ was measured:

$$21,\ 21,\ 23,\ 25,\ 26,\ 27,\ 28,\ 29,\ 29,\ 29,\ 30,\ 30,\ 32,\ 38,\ 38,\ 42,\ 43,\ 44,\ 48,\ 50,\ 76$$



a Determine the median $\mathrm{VO_2Max}$.

b Determine the upper quartile value.

c Determine the lower quartile value.

d Consider the box plot for this data set and state whether the results are positively or negatively skewed.

e State the value of the outlier.

f An average untrained healthy person has a $\mathrm{VO_2Max}$ between 30 and 40. What can we say about the exercise habits of the majority of this group of people?